

Trainings ▾

General Information

Marine Applications

These instructions are for a single unit drive shaft, which attaches to the engine and transmission with flange mount U-joints. Some marine engines are equipped with drive shafts that are various lengths, depending on the boat builder design. This allows the engine to be mounted some distance from the propulsion system drives. The drive system discussed in this procedure is the Pod drive shaft. This procedure will cover removal, alignment, and installation.

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Preparatory Steps

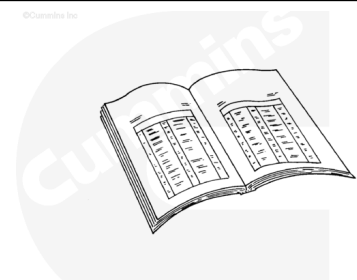
Marine Applications

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)

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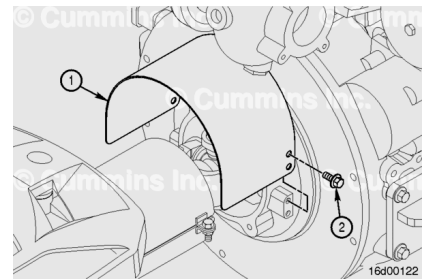
Remove

Marine Applications

Back out the capscrews (2) supporting the engine end shield (1).

Support the shield while removing the last capscrew.

Remove the shield.



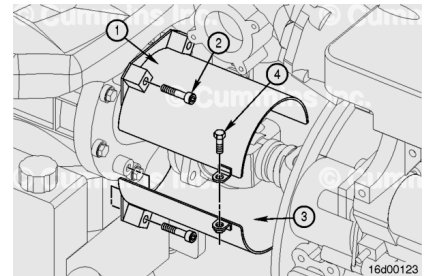
Back out the bolts (4) which are securing the two halves of the transmission end shield together.

Back out the capscrews (2) which are supporting the upper half of the shield (1).

Remove the upper half of the shield.

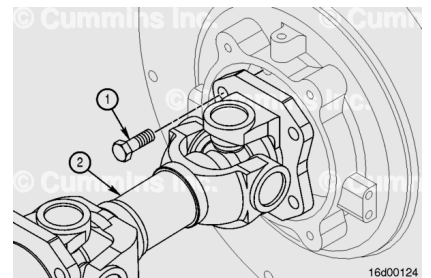
Back out the capscrews, which are supporting the lower half of the shield (3).

Remove the shield.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The longer drive shafts can be very heavy. The technician should consider whether two technicians are needed for the procedure.

Back out the capscrews (1) from each end of the drive shaft (2). Support the shaft while removing the last capscrew from each end. Set the drive shaft to one side.

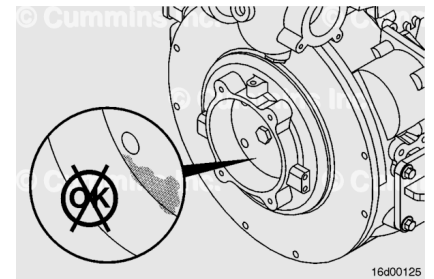
Setup

Marine Applications

Alignment Kit, Part Number 2892196

- Check the engine side coupler flange for proper seating prior to alignment tool installation.

- Check all mating surfaces and magnets for debris which could interfere with tool seating.
- When installing onto a QSM11, install gauge mounting adapter, Part Number 2892182, and alignment tool.

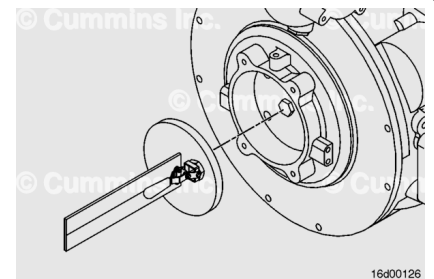


Install alignment tool marked “ENGINE” to the engine drive coupling flange with blade parallel to the ground.

Install alignment tool marked “GEAR” to the transmission input flange with blade parallel to the ground.

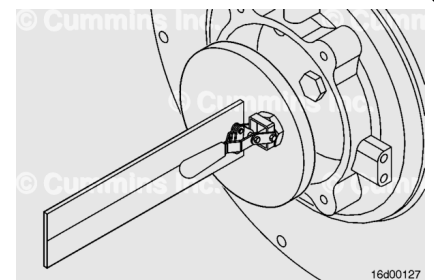
Confirm that the tools are firmly attached with no movement of any kind.

When installing onto a QSM11, install gauge mounting adapter, Part Number 2892182, and alignment tool.

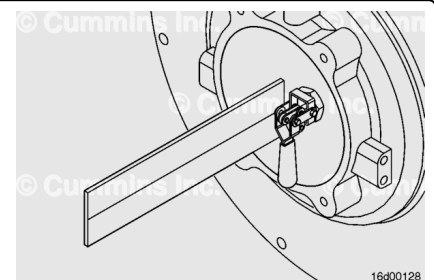


Pull out on the release lever to release the tools for changing position or removal.

Do **not** pry the alignment tools off with a pry bar, screwdriver, or similar tool, because damage can occur.



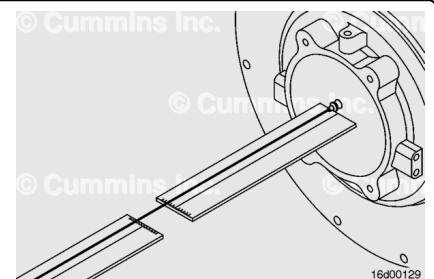
Make sure the release lever is disengaged when measuring.



For driveshafts longer than 254 mm [10 in]:

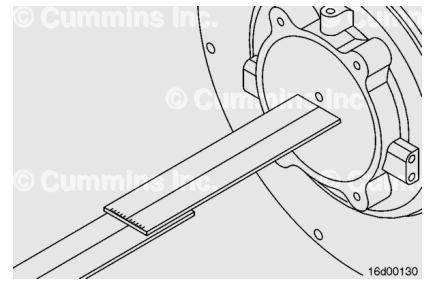
- Insert the indicator line into the holes in the tool blades using the pegs to hold the line tight.
- Install line in the tool so **not** to interfere with free movement of the line.

Make sure the line is tight and does **not** sag.



The alignment tool blades overlap with the 254 mm [10 in] driveshaft installation and no indicator line is necessary.

When installing the alignment tools, confirm that the blades are **not** against each other, and the blades are parallel to the ground.



Measure

Marine Applications

Driveshafts longer than 254 mm [10 in]:

- Check the horizontal alignment of the engine and pod.

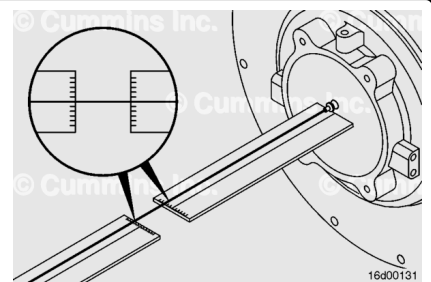
Read the end of the tool where the line crosses the degree scale.

Move the engine side to side to get as close to "0" as possible.

Rotate both tools 180 degrees.

Confirm that the readings in this position are the same as the previous test. If **not**, check the tool fit-up to the flanges again.

Horizontal Alignment: 0 degrees $\pm$ 1/2 degree.

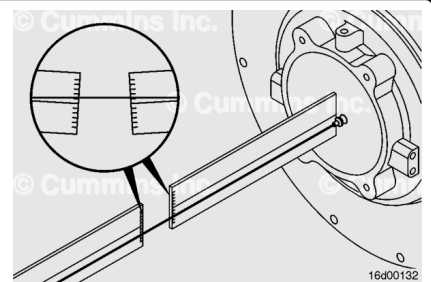


Check the vertical alignment:

- Rotate both tools 90 degrees so they are perpendicular to the ground.

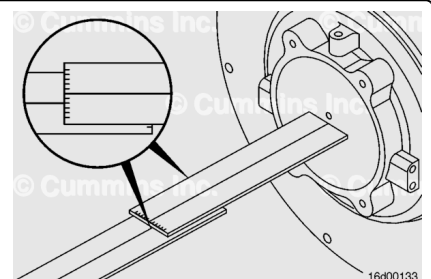
Read the angles of the pod and engine where the lines cross the scales at the free end of the blades. This gives the operator the vertical angles of the engine and pod. The readings on both scales should be very close to the same.

Vertical Alignment: 2 degrees $\pm$ 1 degree.



Drive shafts 254 mm [10 in]:

- Check the horizontal alignment of the pod and engine.
Read the angle where the tool center lines cross the scale at the free end of the blades.



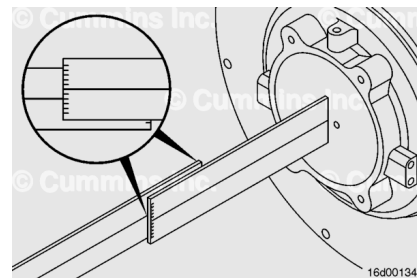
Move the engine side to side to get as close to "0" degrees as possible on the pod and engine ends.

Horizontal Alignment: 0 degrees $\pm$ 1/2 degree.

Rotate both tools 90 degrees so they are perpendicular to the ground.

Check the vertical alignment of the engine and pod:

- Read the angle where the tool center lines cross the scales at the free ends of the blades.



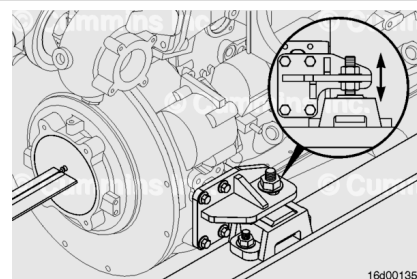
Adjust

Marine Applications

Adjust the engine mounts to get as close to 2 degrees as possible on both the engine and pod ends.

The readings on both scales should be very close to the same.

Vertical alignment: 2 degrees $\pm$ 1 degree.

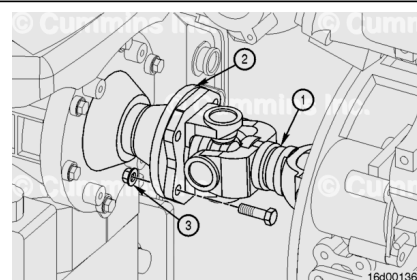


Install

Marine Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The longer driveshafts can be very heavy. The technician should consider whether two technicians are needed for this procedure.

Fully collapse (push) the splined portion of the driveshaft (1) together.

Use flange bolt (3/8-24, 1-1/2 in long). Attach the driveshaft to the transmission input shaft flange (2).

Note : These bolts can not be used on the opposite side of the drive shaft because they will bottom out.

Tighten the eight locknuts (3) on the four transmission input shaft flange bolts.

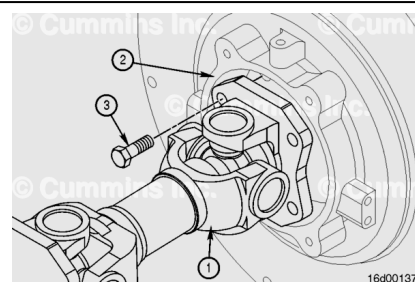
Torque Value: 61 n•m [45 ft-lb]

Attach the driveshaft (1) to the engine coupler (2).

Tighten the eight capscrews (3) on the engine coupler (2).

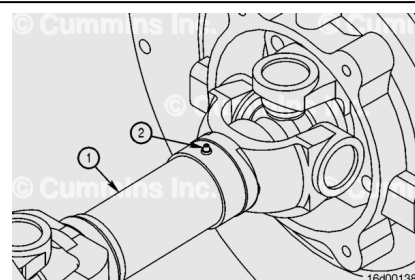
Use the flange bolt (3/8-24, 1 in long).

Torque Value: 61 n•m [45 ft-lb]



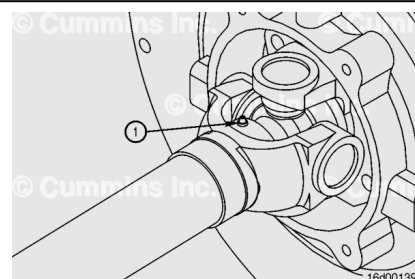
Lubricate the driveshaft slip-joint (1) through the grease fitting (2) by applying approximately three to six pumps of specified grease from a typical hand-operated grease gun.

Use Mercury™ Engine Coupler Spline Grease, Part Number 92-802869A1, or equivalent.



Driveshafts over 254 mm [10 in] have a grease fitting in the cross member of the U joints (1). These should be lubricated with the specified grease.

Use Mercury™ U-joint and Gimbal Bearing Grease, Part Number 92-802870A1, or equivalent.



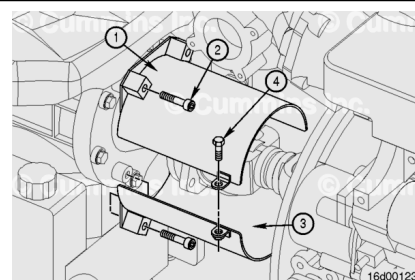
Install the top (1) and bottom (3) driveshaft shields on the transmission end as shown.

Coat the capscrew threads with Loctite™ 277, or equivalent.

Tighten the driveshaft shield retaining capscrews and nuts on the transmission end.

Torque Value:

Socket Head Capscrew (2) 36 n•m [27 ft-lb]



Torque Value:

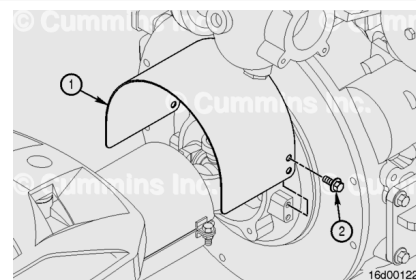
Capscrew and Nut 23 n•m [204 in-lb]

Install the engine end driveshaft shield (1) in the flywheel housing cover.

Coat the capscrew threads with Loctite™ 277, or equivalent.

Tighten the driveshaft shield retaining capscrews (2).

Torque Value: 47 n•m [35 ft-lb]

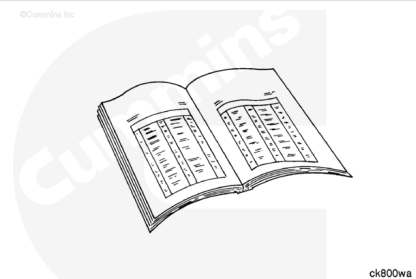


Finishing Steps

Marine Applications

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Start the equipment and check the operation.